

Getting started

First you need to import the relevant modules and create a graphical user interface (GUI). Then you need to create a canvas to draw the shapes on.



You'll shuffle the shapes using the `random` module.

Use `Tkinter` to create the GUI.

1 Create a new file

Open IDLE. Create a new file and save it as "snap.py".

2 Add modules

First import the `random` and `time` modules, and parts of `Tkinter`. `Time` lets you create a delay so that the player is able to read a "SNAP!" or "WRONG!" message before the next shape is shown. `HIDDEN` lets you hide each shape until you want to show it with `NORMAL` – otherwise all the shapes will appear on the screen at the start of the game.

```
import random
import time
from tkinter import Tk, Canvas, HIDDEN, NORMAL
```

3 Set up the GUI

Now type the code shown here to create a `Tkinter` window (also called a root widget) with the title "Snap". Run the code to check it. The window may be hidden behind the other windows on the desktop.

```
from tkinter import Tk, Canvas, HIDDEN, NORMAL
```

```
root = Tk()
root.title('Snap')
```

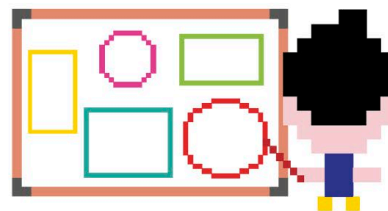
4 Create the canvas

Type this line to create the canvas – the blank space on which the shapes will appear.

```
root.title('Snap')
c = Canvas(root, width=400, height=400)
```

Making the shapes

The next stage is to create the coloured shapes using functions from `Tkinter`'s `Canvas` widget. You'll draw circles, squares, and rectangles, each in four different colours.



5 Make a store for the shapes

You need to make a list so that you can store all the shapes somewhere. Add this line at the bottom of your file.

```
c = Canvas(root, width=400, height=400)
```

```
shapes = []
```